

# Migrating a Database to Amazon RDS

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## Scenario

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The café currently uses a single Amazon Elastic Compute Cloud (Amazon EC2) instance to host its web server, database, and application code.

Meanwhile, the café business has grown. The order history that is stored in the database provides valuable business information that the café staff doesn't want to lose. Martha uses the data for accounting, and Frank looks at it occasionally to plan how many of each dessert type he should bake.

Sofía has additional concerns. The database must be consistently upgraded and patched, and she doesn't always have time to do these tasks. Also, administering the database is a specialized skill. Training others to do database administration isn't something that she wants to spend time on. Meanwhile, Sofía is also concerned that the café isn't doing data backups as often as they should.

Finally, Martha also wants to reduce labor costs that are associated with the technical learning investment that's needed to manage the database.

## Solution

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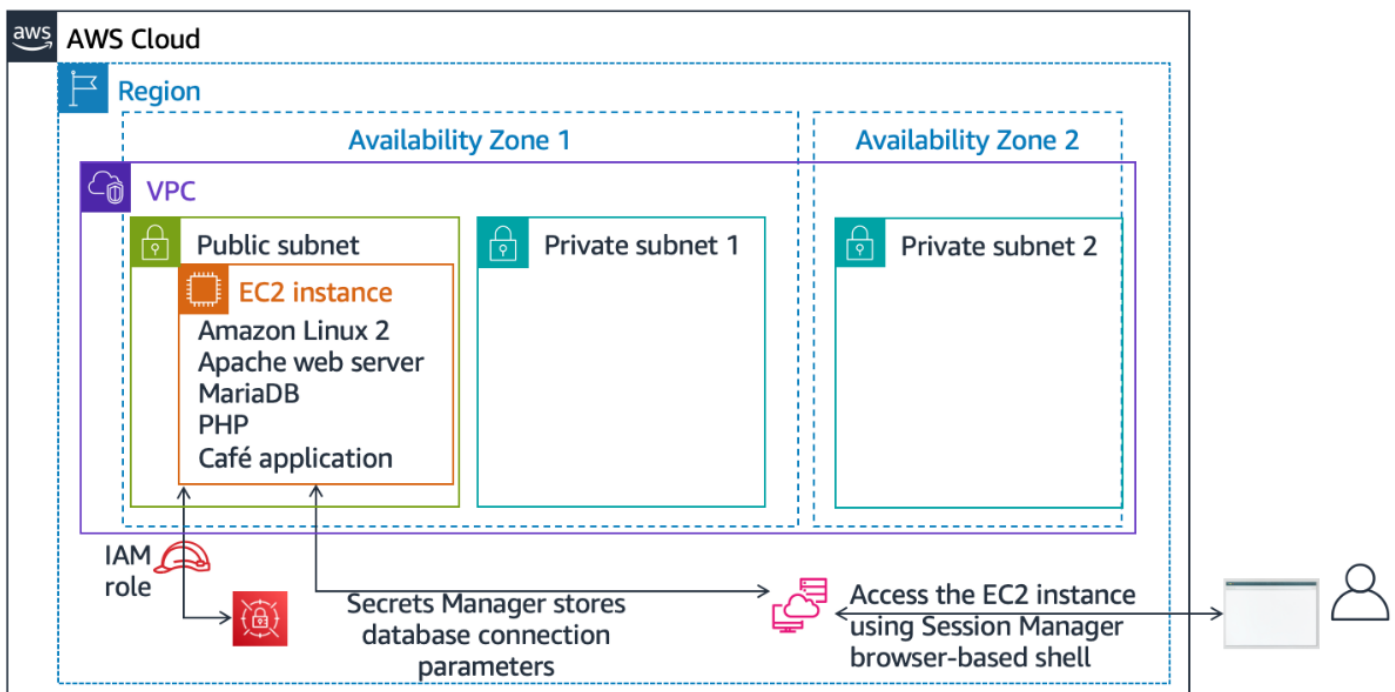
### **Preconfigured Infrastructure**

Before beginning the hands-on portion of the lab, several AWS resources were already provisioned to streamline the migration process. This allowed me to focus on the core objectives of the lab while still reviewing each component to fully understand its role and configuration.

### **Pre-deployed Resources:**

- **VPC (Virtual Private Cloud):** A private network environment within AWS, preconfigured with the following:
  - **Public Subnet:** Hosts the EC2 instance and allows internet access.
  - **Private Subnet 1 & Private Subnet 2:** Used to securely deploy services such as the RDS database, with no direct internet exposure.

- **EC2 Instance:** An Amazon Linux 2 instance with the following components already installed:
  - **Apache Web Server**
  - **PHP**
  - **MariaDB Database**
  - **The Café Web Application**
- **IAM Role:** Attached to the EC2 instance, this role grants secure access to AWS services like **Secrets Manager** and **AWS Systems Manager**, without the need for manual credential configuration.
- **AWS Secrets Manager:** Stores sensitive data such as database credentials, allowing the application to retrieve them securely.
- **AWS Systems Manager – Session Manager:** Enabled to allow secure browser-based access to the EC2 instance without the need for SSH or key pairs.



## 🎯 Lab Focus and Objectives

In this lab, the main focus is on **migrating a database from an EC2 instance to Amazon RDS**. Specifically, we work with a **MariaDB** database that initially runs on an EC2 instance and migrate it to a **managed MariaDB database on Amazon RDS**. Additionally, the café's web application is updated to interact with the new RDS instance, so all future orders are stored in the new environment.

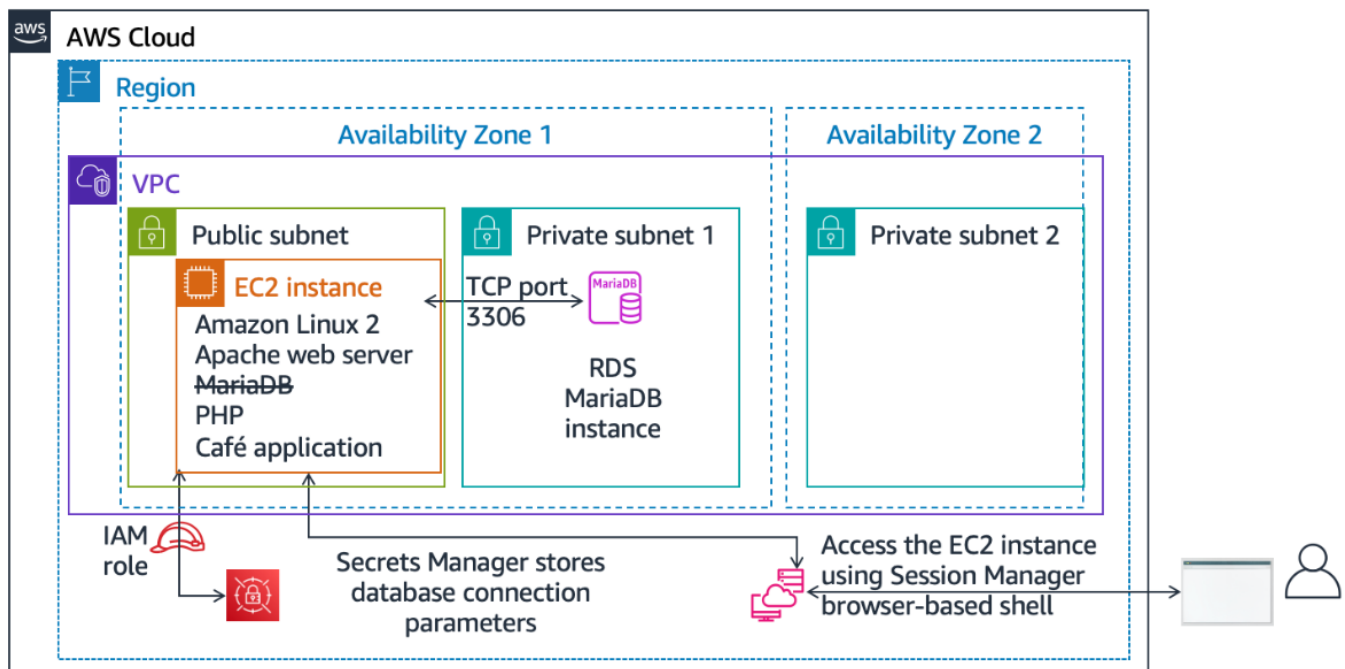
By completing this lab, you will gain hands-on experience with the following key tasks:

- ☒ **Creating an Amazon RDS database instance**
- ☒ **Exporting data from a MariaDB database using `mysqldump`**
- ☒ **Connecting to the RDS database using a SQL client**

- ☒ **Migrating data from the EC2-hosted MariaDB to the new RDS instance**
- ☒ **Reconfiguring the café web application to use the RDS database for all future data storage**

These steps simulate a real-world migration scenario where scalability, manageability, and data security are top priorities.

- At the end of this lab, your architecture will look like the following example:



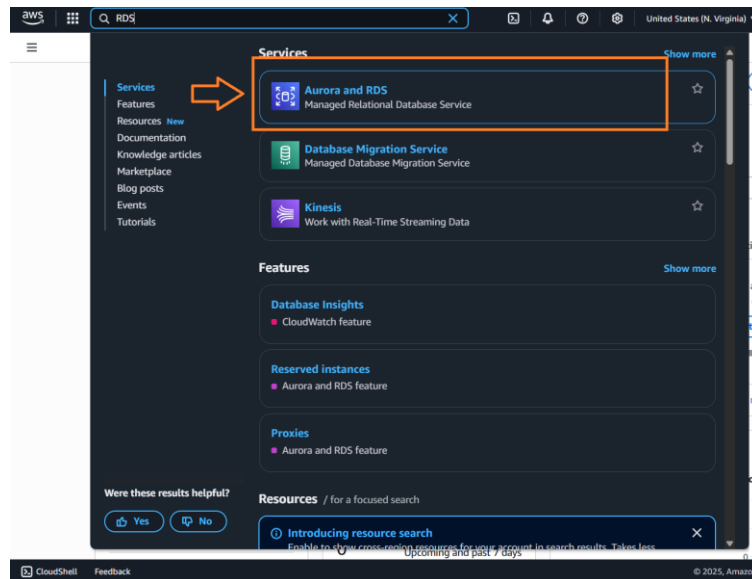
## A business request: Creating an RDS instance for the café application

After a conversation with Olivia—the AWS solutions architect who often comes in for coffee—Sofía decided that the café needs a database solution that is easier to maintain. In addition, the database should provide essential features such as durability, scalability, and high performance.

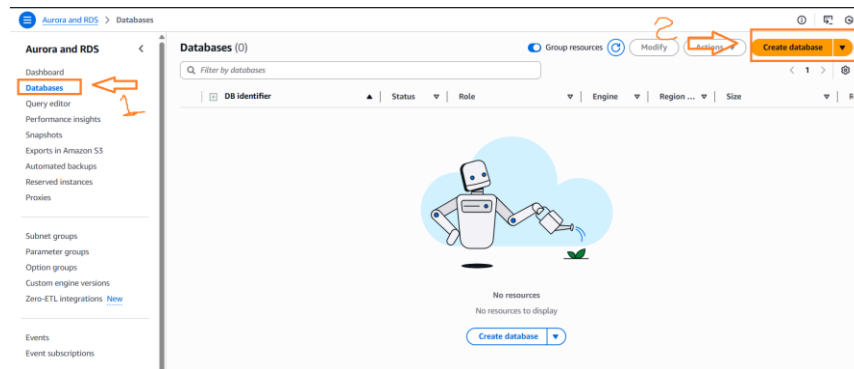
In the first part of this lab, you take on the role of Sofía. You create an RDS instance that the café can use as the data storage layer for the café website. You also connect to the EC2 instance and analyze the details of the café web application.

### ☒ Step 1: Creating an RDS instance

- Go to the Amazon RDS console.



- From the left-hand **navigation bar**, click on **“Databases”**.  
Then, click the **“Create database”** button to start the process of provisioning a new RDS instance.



- **Step 1: Select the Database Engine**
  - In the **Engine options** section:  
For **Engine type**, select **MariaDB**.  
    - \* **MariaDB is a popular open-source relational database that is compatible with MySQL and commonly used for web applications — including the café’s web application running on the EC2 instance, which is also designed to work with a MariaDB database.**

**Choose a database creation method**

☒ **Standard create**  
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ **Easy create**  
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

**Engine options**

**Engine type** [Info](#)

☐ Aurora (MySQL Compatible)

☐ PostgreSQL

☐ Microsoft SQL Server

☐ Aurora (PostgreSQL Compatible)

☒ **MariaDB**

☐ MySQL

☐ Oracle

☐ IBM Db2

## ◆ Step 2: Choose a Template

- Under the **Templates** section:  
Choose **Dev/Test**.

✳️ *This template is optimized for development and testing environments, providing a cost-effective setup.*

**Templates**

Choose a sample template to meet your use case.

☐ **Production**  
Use defaults for high availability and fast, consistent performance.

☒ **Dev/Test**  
This instance is intended for development use outside of a production environment.

☐ **Sandbox**  
To develop new applications, test existing applications, or gain hands-on experience with Amazon RDS.

## ◆ Step 3: Configure Database Settings

- In the **Settings** section, fill in the following:

**DB instance identifier:** CafeDatabase

**Master username:** admin

**Master password and Confirm password:** Caf3DbPassw0rd!

✳️ *Use the exact password provided. It's recommended to copy and paste it to avoid typos.*

**Settings**

**DB instance identifier** [Info](#)  
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.  
CafeDatabase

**▼ Credentials Settings**

**Master username** [Info](#)  
Type a login ID for the master user of your DB instance.  
admin

**Credentials management**  
You can use AWS Secrets Manager or manage your master user credentials.  
☐ Managed in AWS Secrets Manager - most secure  
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.  
☒ **Self managed**  
Create your own password or have RDS create a password that you manage.

**Master password** [Info](#)  
Password strength: Strong  
Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / " ' @

**Confirm master password** [Info](#)

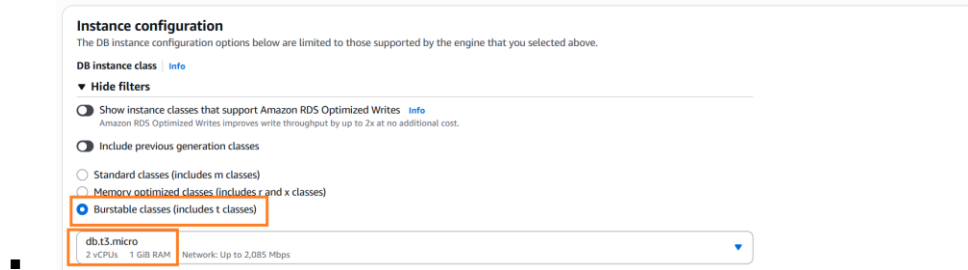
- ◆ **Step 4: Configure Instance Class**

- In the **Instance configuration** section:

For **DB instance class**, select **Burstable classes (includes t classes)**.

Then choose **db.t3.micro**.

✳️ *This instance class is suitable for small workloads and is eligible for the AWS Free Tier.*



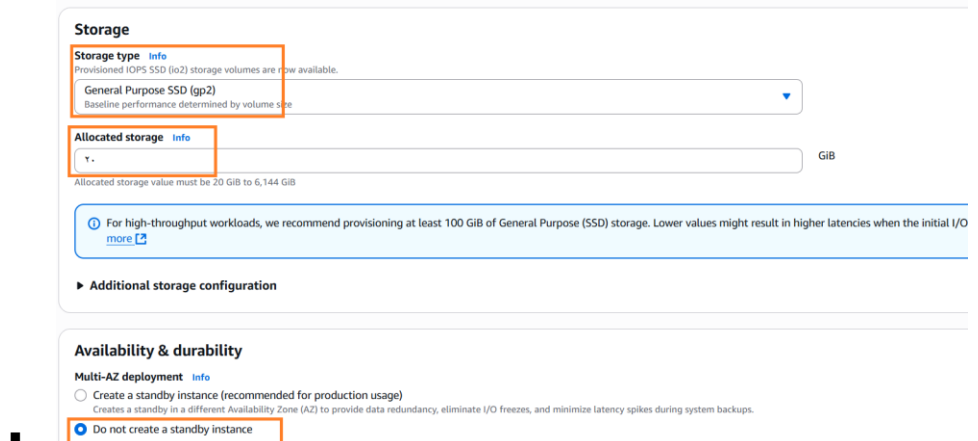
- ◆ **Step 5: Configure Storage**

- In the **Storage** section:

**Storage type:** Select **General Purpose SSD (gp2)**

**Allocated storage:** Enter **20 GiB**

✳️ *This provides sufficient performance for lab purposes and reduces cost.*



- ◆ **Step 6: Configure Availability and Durability**

- In the **Availability & durability** section:

Choose **Do not create a standby instance**

✳️ *A standby instance isn't necessary for this lab scenario since high availability isn't a focus here.*

#### Availability & durability

##### Multi-AZ deployment [Info](#)

☐ Create a standby instance (recommended for production usage)

Creates a standby in a different Availability Zone (AZ) to provide data redundancy, eliminate I/O freezes, and minimize latency spikes during system backups.

☒ Do not create a standby instance

## • Step 7: Configure Connectivity

- In the **Connectivity** section:

**Virtual private cloud (VPC):** Select **Lab VPC**

**DB subnet group:** Select **lab-db-subnet-group**

**Public access:** Choose **No**

**Existing VPC security group:** Select **dbSG** and uncheck the **default security group**

**Availability Zone:** Select the first one ending in a (e.g., us-east-1a)

**Database port:** Leave as default 3306

✳ *These settings ensure the RDS instance is securely placed in a private subnet inside the lab's VPC.*

**Connectivity** [Info](#)

**Compute resource**  
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ Don't connect to an EC2 compute resource  
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ Connect to an EC2 compute resource  
Set up a connection to an EC2 compute resource for this database.

**Network type** [Info](#)  
To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

☒ IPv4  
Your resources can communicate only over the IPv4 addressing protocol.

☐ Dual-stack mode  
Your resources can communicate over IPv4, IPv6, or both.

**Virtual private cloud (VPC)** [Info](#)  
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Lab VPC (vpc-0978aedb0c58cd1ed)  
3 Subnets, 2 Availability Zones

Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

**DB subnet group** [Info](#)  
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

lab-db-subnet-group  
2 Subnets, 2 Availability Zones

**Public access** [Info](#)

☐ Yes  
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups which resources can connect to the database.

☒ No  
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

- At this point, the dbSG security group does not have any inbound or outbound rules configured.

We will add the necessary rules later to allow the required connectivity to the RDS instance.

**Existing VPC security groups**  
Choose one or more options

dbSG

c170897a4412341111070541t1w746511720335-CafeSG-rqYrPdIRBuAB

default

## ▼ Additional configuration

### Database port [Info](#)

TCP/IP port that the database will use for application connections.

3306

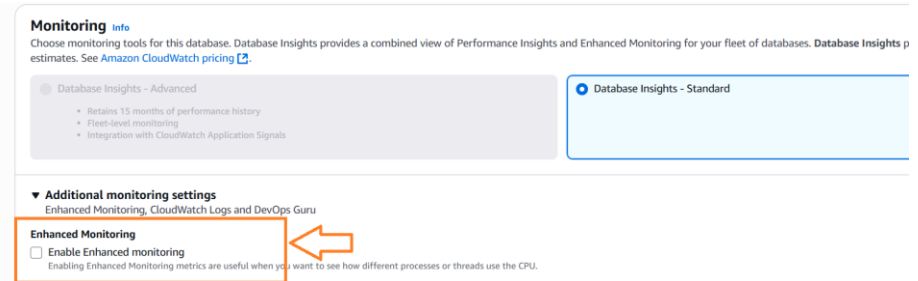


## • ♦ Step 8: Disable Enhanced Monitoring

- In the **Monitoring** section:

Uncheck **Enable Enhanced Monitoring**

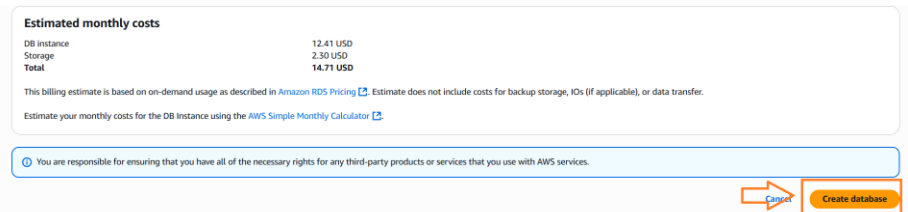
✳ *Enhanced monitoring is not supported in the lab environment, so this feature should remain disabled.*



## • ♦ Step 9: Create the Database

- Scroll to the bottom and click on **Create database**.

✳ *This initiates the provisioning of your RDS instance with the configuration you specified.*



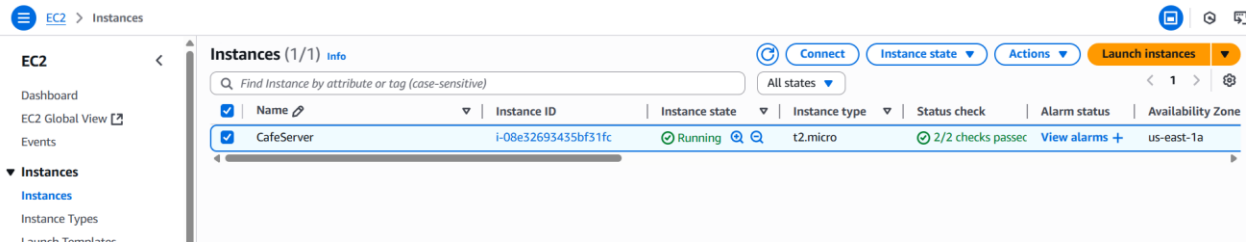
- The creation process of the RDS database instance might take a few minutes to complete. While the database is being provisioned, we'll switch over to the EC2 instance to explore its configuration and examine the components of the café web application.

## ☑ Step 2: Analyzing the existing café application deployment

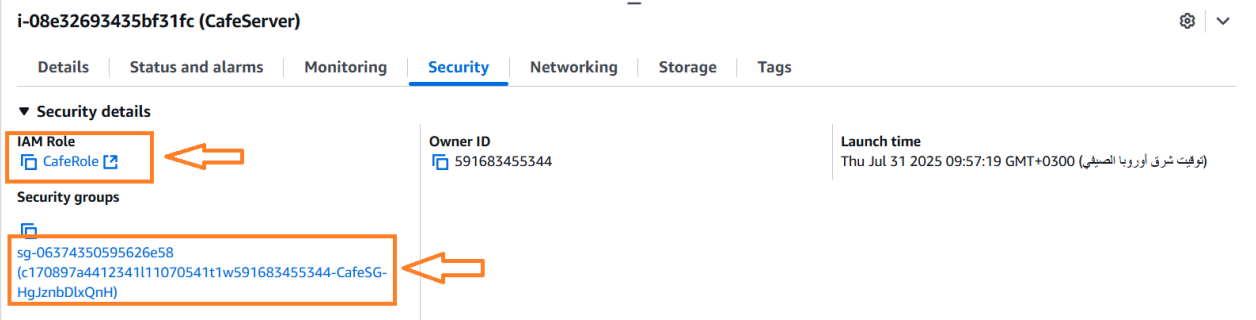
- **Next, navigate to the EC2 dashboard and view the list of instances.**

You will find one instance named **CafeServer**, which we mentioned earlier. This instance hosts the web server for the café application.

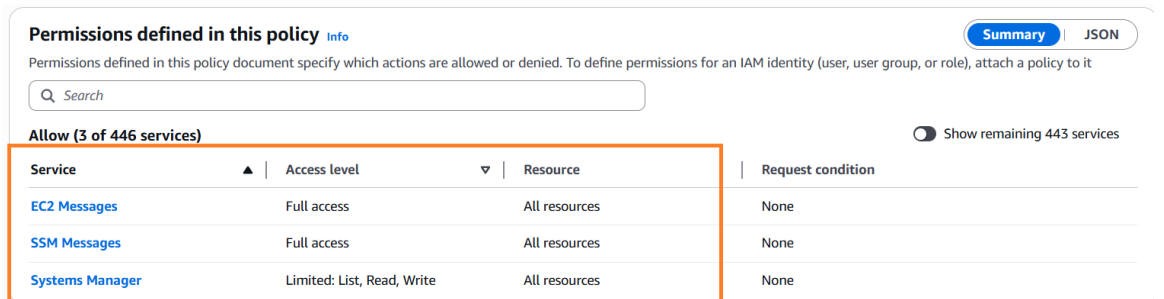




- In the **Security** section of the instance details, you'll see that the instance is associated with a security group and has an IAM role attached to it.



- After inspecting the permissions attached to the **CafeRole**, we found that it includes two customer-created inline policies alongside the AWS managed policy:
  - **AmazonSSMManagedInstanceCore (AWS Managed):**  
This managed policy allows the instance to be managed using AWS Systems Manager (SSM), enabling features like Session Manager and Run Command. We'll dive deeper into how SSM works later in the lab.



- **importkeyAndSsmSm (Customer Inline):**  
This custom policy grants the instance permission to import key pairs into EC2. This is particularly useful when you want to securely manage SSH access by uploading public keys directly through the AWS Management Console or CLI.

## Modify permissions in importkeyAndSsmSm [Info](#)

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

### Policy editor

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Action": "ec2:ImportKeyPair",
6       "Resource": "*",
7       "Effect": "Allow"
8     }
9   ]
10 }
```

- **root (Customer Inline):**

This inline policy provides permissions for interacting with AWS Secrets Manager. Specifically, it allows the instance to create, retrieve, describe, list, update, and store secret values within the account. These permissions are essential for applications that store sensitive credentials (like database passwords or API keys) securely in Secrets Manager and need to access them at runtime.

## Modify permissions in root [Info](#)

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

### Policy editor

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Action": [
6         "secretsmanager:CreateSecret",
7         "secretsmanager:GetSecretValue",
8         "secretsmanager:DescribeSecret",
9         "secretsmanager:PutSecretValue",
10        "secretsmanager:ListSecrets",
11        "secretsmanager:UpdateSecret"
12      ],
13       "Resource": "arn:aws:secretsmanager:*:591683455344:secret:*",
14       "Effect": "Allow"
15     }
16   ]
17 }
```

- And if we check **AWS Secrets Manager**, we'll find the **database secrets** used by the web application running on the EC2 instance. As mentioned earlier, there's a script running on the instance that handles creating and managing these secrets automatically — the script is called `set-app-parameters.sh` and is located inside the `setup` folder of the project.

**Secrets** Store a new secret

Filter secrets by name, description, tag key, tag value, owning service or primary Region

| Secret name          | Description | Last retrieved (UTC) |
|----------------------|-------------|----------------------|
| /cafe/dbPassword     | -           | July 31, 2025        |
| /cafe/dbUser         | -           | July 31, 2025        |
| /cafe/dbName         | -           | July 31, 2025        |
| /cafe/dbUrl          | -           | July 31, 2025        |
| /cafe/currency       | -           | July 31, 2025        |
| /cafe/timeZone       | -           | July 31, 2025        |
| /cafe/showServerInfo | -           | July 31, 2025        |

```

1 #!/bin/bash
2 #
3 # Script to set the parameters for the cafe application in the Parameter Store.
4 #
5 # Get the region where the instance is running, and set it as the default AWS region.
6 # This ensures that we are using the Parameter Store in the region where the instance is running.
7
8 echo "Setting the default AWS region..."
9 TOKEN=$(curl -s PUT "http://169.254.169.254/latest/api/token" -H "X-aws-ec2-metadata-token-ttl-seconds: 21600")
10 curl -H "X-aws-ec2-metadata-token: $TOKEN" -v http://169.254.169.254/latest/meta-data/
11
12 ar=$(curl -H "X-aws-ec2-metadata-token: $TOKEN" -v http://169.254.169.254/latest/meta-data/placement/availability-zone)
13 region=$(echo $ar | sed 's/-$//')
14 export AWS_DEFAULT_REGION=$region
15 echo "Region = $AWS_DEFAULT_REGION"
16
17 # Set the application parameter values.
18 publicDNS=$(curl -H "X-aws-ec2-metadata-token: $TOKEN" -v http://169.254.169.254/latest/meta-data/public-hostname)
19 echo "Public DNS = $publicDNS"
20
21 echo "Setting the application parameter values in the Secrets Manager..."
22 aws secretsmanager create-secret --name "/cafe/showServerInfo" --secret-string "false"
23 aws secretsmanager create-secret --name "/cafe/timeZone" --secret-string "America/New_York"
24 aws secretsmanager create-secret --name "/cafe/currency" --secret-string "USD"
25 aws secretsmanager create-secret --name "/cafe/dbUrl" --secret-string "publicDNS"
26 aws secretsmanager create-secret --name "/cafe/dbName" --secret-string "cafe_db"
27 aws secretsmanager create-secret --name "/cafe/dbUser" --secret-string "admin"
28 aws secretsmanager create-secret --name "/cafe/dbPassword" --secret-string "1mb1234"
29
30 echo "Application Secrets Setup script completed."
31
  
```

- If we check the Security Group settings for the EC2 instance, we'll see that it allows **HTTP** and **SSH** inbound traffic from the public internet (0.0.0.0/0). This means the instance is accessible from any IP address, which can pose a potential security risk if not properly managed.

**sg-06374350595626e58 - c170897a441234111070541t1w591683455344-CafeSG-HgJznBDlxQnH** Actions

**Details**

**Security group name**  
c170897a441234111070541t1w591683455344-CafeSG-HgJznBDlxQnH

**Owner**  
591683455344

**Security group ID**  
sg-06374350595626e58

**Inbound rules count**  
2 Permission entries

**Description**  
Enable SSH, HTTP access

**Outbound rules count**  
1 Permission entry

**VPC ID**  
vpc-08ee09c849ac25653

[Inbound rules](#) | [Outbound rules](#) | [Sharing - new](#) | [VPC associations - new](#) | [Tags](#)

**Inbound rules (2)** Manage tags Edit inbound rules

| Security group rule ID | IP version | Type | Protocol | Port range | Source    |
|------------------------|------------|------|----------|------------|-----------|
| sgr-03127def5f426bd33  | IPv4       | HTTP | TCP      | 80         | 0.0.0.0/0 |
| sgr-0030b205f897131a2  | IPv4       | SSH  | TCP      | 22         | 0.0.0.0/0 |

- Now, let's test the web application hosted on the EC2 instance. We'll retrieve the **Public IP address** or **Public DNS name**, and use it in the browser to access the website — for example: <http://ec2-44-202-147-178.compute-1.amazonaws.com/cafe/>.

# Café

[Home](#) [About Us](#) [Contact Us](#) [Menu](#) [Order History](#)



Our café offers an assortment of delicious and delectable pastries and coffees that will put a smile on your face. From cookies to croissants, tarts and cakes, each treat is especially prepared to excite your tastebuds and brighten your day!

Frank bakes a rich variety of cookies. Try them all!



Tea,  
Coffee,  
Lattes,  
and Hot  
Chocolate.



Our tarts are always  
a customer favorite!



- Now let's try placing an order and submitting it to see how the **Order History** gets updated.

# Café

[Home](#) [Menu](#) [Order History](#)

## Pastries



### Croissant

\$1.50

Fresh, buttery and fluffy... Simply delicious!

Quantity:



### Donut

\$1.00

We have more than half-a-dozen flavors!

Quantity:



### Chocolate Chip Cookie

\$2.50

Made with Swiss chocolate with a touch of Madagascar vanilla

Quantity:



**Muffin**  
\$3.00  
Banana bread, blueberry, cranberry or apple  
Quantity:



**Strawberry Blueberry Tart**  
\$3.50  
Bursting with the taste and aroma of fresh fruit  
Quantity:



**Strawberry Tart**  
\$3.50  
Made with fresh ripe strawberries and a delicious whipped cream  
Quantity:

Drinks



**Coffee**  
\$3.00  
Freshly-ground black or blended Columbian coffee  
Quantity:



**Hot Chocolate**  
\$3.00  
Rich and creamy, and made with real chocolate  
Quantity:



**Latte**  
\$3.50  
Offered hot or cold and in various delicious flavors  
Quantity:

Order Total: \$5.50

Submit Order

Reset Order

○

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# Café

Home Menu Order History

## Order Confirmation

Thank for your order! It will be available for pickup within 15 minutes. Your order number and details are shown below.

Order Number: 25 Date: 2025-07-31 Time: 08:16:36 Total Amount: \$5.50

| Item                  | Price  | Quantity | Amount |
|-----------------------|--------|----------|--------|
| Donut                 | \$1.00 | 3        | \$3.00 |
| Chocolate Chip Cookie | \$2.50 | 1        | \$2.50 |

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- And if we take a look at the **Order History**, we'll see the order we just placed along with some older orders that were already stored in the database.

# Café

| Home                      |  |                  |                |                       | Menu    | Order History |
|---------------------------|--|------------------|----------------|-----------------------|---------|---------------|
| Order History             |  |                  |                |                       |         |               |
| Order Number: 25          |  | Date: 2025-07-31 | Time: 08:16:36 | Total Amount: \$5.50  |         |               |
| Item                      |  |                  | Price          | Quantity              | Amount  |               |
| Donut                     |  |                  | \$1.00         | 3                     | \$3.00  |               |
| Chocolate Chip Cookie     |  |                  | \$2.50         | 1                     | \$2.50  |               |
|                           |  |                  |                |                       |         |               |
| Order Number: 24          |  | Date: 2020-07-28 | Time: 13:14:07 | Total Amount: \$35.00 |         |               |
| Item                      |  |                  | Price          | Quantity              | Amount  |               |
| Strawberry Blueberry Tart |  |                  | \$3.50         | 4                     | \$14.00 |               |
| Strawberry Tart           |  |                  | \$3.50         | 3                     | \$10.50 |               |
| Latte                     |  |                  | \$3.50         | 3                     | \$10.50 |               |
|                           |  |                  |                |                       |         |               |
| Order Number: 23          |  | Date: 2020-07-28 | Time: 13:13:54 | Total Amount: \$6.00  |         |               |
| Item                      |  | Price            | Quantity       |                       | Amount  |               |
| Coffee                    |  | \$3.00           | 2              |                       | \$6.00  |               |
|                           |  |                  |                |                       |         |               |
| Order Number: 22          |  | Date: 2020-07-21 | Time: 13:13:47 | Total Amount: \$33.50 |         |               |
| Item                      |  |                  | Price          | Quantity              | Amount  |               |
| Chocolate Chip Cookie     |  |                  | \$2.50         | 3                     | \$7.50  |               |
| Strawberry Blueberry Tart |  |                  | \$3.50         | 4                     | \$14.00 |               |

- **Accessing the EC2 Instance via AWS Systems Manager (SSM)**

- Next, we'll connect to the EC2 instance using **AWS Systems Manager**, which allows us to open a terminal session **directly in the browser**—without needing SSH or key pairs.

-  **What is AWS Systems Manager?**

- **AWS Systems Manager (SSM)** is a management service that helps you securely access, configure, and automate operational tasks on your EC2 instances, on-premises servers, and virtual machines. One of its key features is **Session Manager**, which allows you to securely connect to EC2 instances using the AWS console or CLI **without exposing the instance to the internet** or managing SSH keys.
- This works by using an **SSM Agent** installed on the EC2 instance, which communicates with the Systems Manager service. To use it, the instance must have:
  - The **SSM Agent installed and running** (enabled by default on Amazon Linux 2 and Ubuntu AMIs),
  - Internet access (or access to SSM endpoints via VPC endpoints),
  - An **IAM role attached to the instance** with the necessary SSM permissions.

-  **Steps to Connect:**

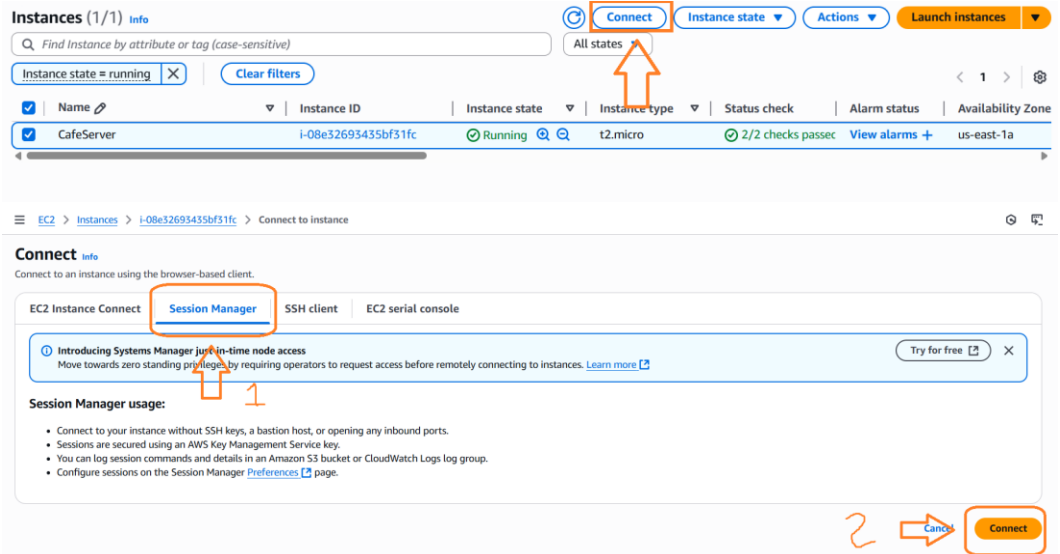
- **Go to the Amazon EC2 Console**

From the AWS Console, navigate to **EC2**, and in the left navigation pane, click **Instances**.

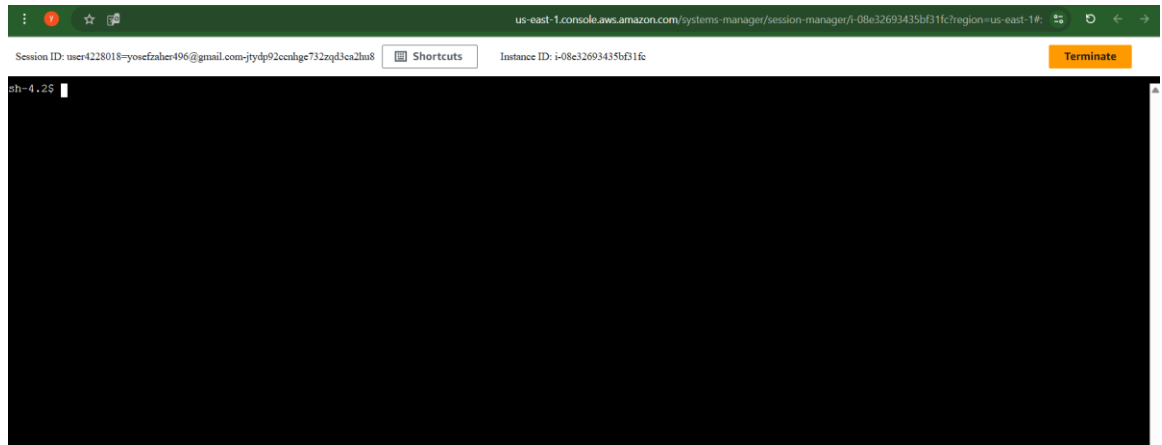
- **Select the CafeServer Instance**

Find the instance named CafeServer, which is running your web application, and select it.

- **Open the Connect Menu**  
Click the **Connect** button at the top.
- **Choose Session Manager**  
In the "Connect to instance" tab, select the **Session Manager** option.
- **Click Connect**  
Finally, click **Connect** to open a terminal session.



- A new browser tab will open with a terminal session **securely connected to your EC2 instance**, allowing you to execute shell commands directly without needing SSH.



- Once connected to the EC2 instance through **AWS Systems Manager Session Manager**, we can begin executing commands in the terminal. Start by entering the following:

```
bash
sudo su
su ec2-user
whoami
cd /home/ec2-user/
```

- - **Analysis of the Commands:**
    - **bash**: Starts a new Bash shell session.



- `sudo su`: Switches to the root user account, granting full administrative access.
- `su ec2-user`: Switches from the root user to the ec2-user account.
- `whoami`: Confirms the currently active user account, which should return ec2-user.
- `cd /home/ec2-user/`: Navigates to the home directory of the ec2-user.
- These steps ensure you're working from the correct environment and user context when managing or interacting with your application.

```
sh-4.2$ bash
[ssm-user@cafeserver bin]$ sudo su
[root@cafeserver bin]# su ec2-user
[ec2-user@cafeserver bin]$ whoami
ec2-user
[ec2-user@cafeserver bin]$ cd /home/ec2-user/
[ec2-user@cafeserver ~]$
```

-  Now that we've reviewed the EC2 instance and verified that everything is configured correctly, we successfully accessed it using **AWS Systems Manager Session Manager**.

## New business requirement: Exporting data from the old database and establishing a connection to the new database

---

Now that you have created a new RDS instance, you can move on to the next step in the café's database migration plan. Next, you export the data from the database that the café application currently uses. You also establish a network connection from the EC2 instance (where the application runs) to the new RDS database instance.



## ☑ Step 3: Working with the database on the EC2 instance

### 🔍 Observing and Exporting MariaDB Data from the EC2 Instance

- In this Step, we observe the configuration of the MariaDB database running on the EC2 instance, and export the existing café order history using the `mysqldump` utility.

### ✅ Step 1: Verify MariaDB Status and Version

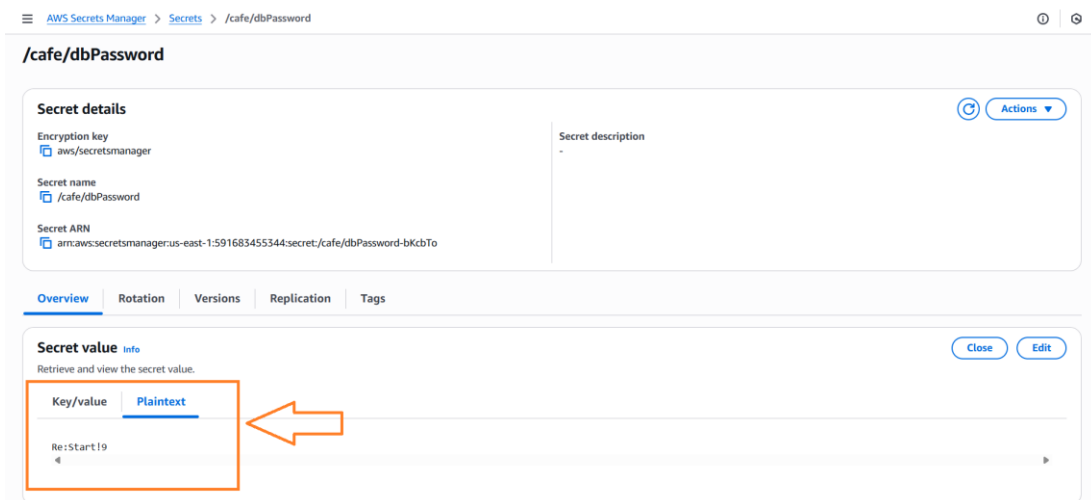
- In the **AWS Systems Manager terminal session**, run the following commands to confirm that the database is running and to check its version:

```
service mariadb status
mysql --version
```

```
[ec2-user@cafeserver ~]$ service mariadb status
Redirecting to /bin/systemctl status mariadb.service
● mariadb.service - MariaDB 10.2 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; vendor preset: disabled)
   Drop-In: /usr/lib/systemd/system/mariadb.service.d
            └─tokudb.conf
   Active: active (running) since Thu 2025-07-31 06:58:48 UTC; 1h 59min ago
     Process: 3720 ExecStartPost=/usr/libexec/mysql-check-upgrade (code=exited, status=0/SUCCESS)
     Process: 3516 ExecStartPre=/usr/libexec/mysql-prepare-db-dir %n (code=exited, status=0/SUCCESS)
     Process: 3492 ExecStartPre=/usr/libexec/mysql-check-socket (code=exited, status=0/SUCCESS)
   Main PID: 3649 (mysqld)
   Status: "Taking your SQL requests now..."
   CGroup: /system.slice/mariadb.service
           └─3649 /usr/libexec/mysqld --basedir=/usr

Jul 31 06:58:48 ip-10-0-0-123.ec2.internal mysql-check-upgrade[3720]: ERROR: ld.so: object '/usr/lib64/libjemalloc.so.1' from LD_PRELOAD cannot be preloaded (can
Jul 31 06:58:48 ip-10-0-0-123.ec2.internal mysql-check-upgrade[3720]: ERROR: ld.so: object '/usr/lib64/libjemalloc.so.1' from LD_PRELOAD cannot be preloaded (can
Jul 31 06:58:48 ip-10-0-0-123.ec2.internal mysql-check-upgrade[3720]: ERROR: ld.so: object '/usr/lib64/libjemalloc.so.1' from LD_PRELOAD cannot be preloaded (can
Jul 31 06:58:48 ip-10-0-0-123.ec2.internal mysql-check-upgrade[3720]: ERROR: ld.so: object '/usr/lib64/libjemalloc.so.1' from LD_PRELOAD cannot be preloaded (can
Jul 31 06:58:48 ip-10-0-0-123.ec2.internal mysql-check-upgrade[3720]: ERROR: ld.so: object '/usr/lib64/libjemalloc.so.1' from LD_PRELOAD cannot be preloaded (can
Jul 31 06:58:48 ip-10-0-0-123.ec2.internal mysql-check-upgrade[3720]: ERROR: ld.so: object '/usr/lib64/libjemalloc.so.1' from LD_PRELOAD cannot be preloaded (can
Jul 31 06:58:48 ip-10-0-0-123.ec2.internal mysql-check-upgrade[3720]: ERROR: ld.so: object '/usr/lib64/libjemalloc.so.1' from LD_PRELOAD cannot be preloaded (can
Jul 31 06:58:48 ip-10-0-0-123.ec2.internal mysql-check-upgrade[3720]: ERROR: ld.so: object '/usr/lib64/libjemalloc.so.1' from LD_PRELOAD cannot be preloaded (can
Jul 31 06:58:48 ip-10-0-0-123.ec2.internal mysql-check-upgrade[3720]: ERROR: ld.so: object '/usr/lib64/libjemalloc.so.1' from LD_PRELOAD cannot be preloaded (can
Jul 31 06:58:48 ip-10-0-0-123.ec2.internal systemd[1]: Started MariaDB 10.2 database server.
Hint: Some lines were ellipsized, use -l to show in full.
[ec2-user@cafeserver ~]$ mysql --version
mysql Ver 15.1 Distrib 10.2.38-MariaDB, for Linux (x86_64) using EditLine wrapper
[ec2-user@cafeserver ~]$
```

### ✅ Step 2: Retrieve Database Password from AWS Secrets Manager



### ✅ Step 3: Connect to MariaDB

- In the **same terminal tab**, connect to the database using the following command:

```
mysql -u root -p
```

- 🖱️ When prompted, paste the password you copied from Secrets Manager.

If the connection is successful, you'll see the MariaDB> prompt.

```
[ec2-user@cafeserver ~]$ mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 14
Server version: 10.2.38-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

-  **Step 4: Explore Existing Database and Data**

- Once connected, run the following commands to inspect the database structure and data:

```
show databases;
use cafe_db;
show tables;
select * from `order`;
```

```
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| cafe_db  |
| information_schema |
| mysql    |
| performance_schema |
| test     |
+-----+
5 rows in set (0.00 sec)

MariaDB [(none)]> use cafe_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [cafe_db]> show tables;
+-----+
| Tables_in_cafe_db |
+-----+
| order              |
| order_item         |
| product            |
| product_group      |
+-----+
4 rows in set (0.00 sec)
```

```
MariaDB [cafe_db]> select * from `order`;
```

| order_number | order_date_time     | amount |
|--------------|---------------------|--------|
| 1            | 2020-06-20 13:09:07 | 20.00  |
| 2            | 2020-06-21 13:09:23 | 24.00  |
| 3            | 2020-06-22 13:09:38 | 32.50  |
| 4            | 2020-06-22 13:09:49 | 10.50  |
| 5            | 2020-06-24 13:10:02 | 42.00  |
| 6            | 2020-06-25 13:10:20 | 14.00  |
| 7            | 2020-06-25 13:10:30 | 35.00  |
| 8            | 2020-06-27 13:10:36 | 6.00   |
| 9            | 2020-06-28 13:10:46 | 18.00  |
| 10           | 2020-07-02 13:10:58 | 18.00  |
| 11           | 2020-07-03 13:11:08 | 19.50  |
| 12           | 2020-07-04 13:11:17 | 9.00   |
| 13           | 2020-07-05 13:11:27 | 26.50  |
| 14           | 2020-07-05 13:11:40 | 42.00  |
| 15           | 2020-07-06 13:12:27 | 35.00  |
| 16           | 2020-07-09 13:12:35 | 19.50  |
| 17           | 2020-07-12 13:12:48 | 57.50  |
| 18           | 2020-07-14 13:13:04 | 34.00  |
| 19           | 2020-07-15 13:13:17 | 29.00  |
| 20           | 2020-07-18 13:13:27 | 14.00  |
| 21           | 2020-07-20 13:13:36 | 17.50  |
| 22           | 2020-07-21 13:13:47 | 33.50  |
| 23           | 2020-07-28 13:13:54 | 6.00   |
| 24           | 2020-07-28 13:14:07 | 35.00  |
| 25           | 2025-07-31 08:16:36 | 5.50   |

```
25 rows in set (0.00 sec)
```

■ MariaDB [cafe\_db]>

- To view the order line items:

```
select * from `order_item`;
```

```
MariaDB [cafe_db]> select * from `order_item`;
```

| order_number | order_item_number | product_id | quantity | amount |
|--------------|-------------------|------------|----------|--------|
| 1            | 1                 | 4          | 6.00     |        |
| 1            | 2                 | 9          | 14.00    |        |
| 2            | 1                 | 2          | 12       | 12.00  |
| 2            | 2                 | 7          | 4        | 12.00  |
| 3            | 1                 | 5          | 5        | 17.50  |
| 3            | 2                 | 7          | 3        | 9.00   |
| 3            | 3                 | 8          | 2        | 6.00   |
| 4            | 1                 | 3          | 3        | 7.50   |
| 4            | 2                 | 8          | 1        | 3.00   |
| 5            | 1                 | 1          | 4        | 6.00   |
| 5            | 2                 | 4          | 4        | 12.00  |
| 5            | 3                 | 7          | 8        | 24.00  |
| 6            | 1                 | 9          | 4        | 14.00  |
| 7            | 1                 | 6          | 10       | 35.00  |
| 8            | 1                 | 7          | 2        | 6.00   |
| 9            | 1                 | 1          | 4        | 6.00   |
| 9            | 2                 | 7          | 4        | 12.00  |
| 10           | 1                 | 2          | 6        | 6.00   |
| 10           | 2                 | 7          | 2        | 6.00   |
| 10           | 3                 | 8          | 2        | 6.00   |
| 11           | 1                 | 5          | 1        | 3.50   |
| 11           | 2                 | 6          | 2        | 7.00   |
| 11           | 3                 | 8          | 3        | 9.00   |
| 12           | 1                 | 1          | 2        | 3.00   |
| 12           | 2                 | 7          | 2        | 6.00   |
| 13           | 1                 | 3          | 4        | 10.00  |
| 13           | 2                 | 5          | 3        | 10.50  |
| 13           | 3                 | 8          | 2        | 6.00   |
| 14           | 1                 | 6          | 8        | 28.00  |
| 14           | 2                 | 9          | 4        | 14.00  |
| 15           | 1                 | 6          | 10       | 35.00  |
| 16           | 1                 | 5          | 3        | 10.50  |
| 16           | 2                 | 8          | 3        | 9.00   |
| 17           | 1                 | 1          | 15       | 22.50  |
| 17           | 2                 | 5          | 10       | 35.00  |
| 18           | 1                 | 3          | 8        | 20.00  |

- **✓ Step 5: Exit the SQL Client**

- When done reviewing the data, exit the MariaDB prompt by typing:

```
exit;
```

```
MariaDB [cafe_db]> exit;  
Bye  
[ec2-user@cafeserver ~]$
```

- **✓ Step 6: Export the Database Using mysqldump**

- To back up the database into a `.sql` file, run the following command:

```
mysqldump --databases cafe_db -u root -p > CafeDbDump.sql
```

-  When prompted, paste the password again from Secrets Manager.
- To confirm the dump file was created:

```
ls
```

```
[ec2-user@cafeserver ~]$ mysqldump --databases cafe_db -u root -p > CafeDbDump.sql  
Enter password:  
[ec2-user@cafeserver ~]$ ls  
CafeDbDump.sql  
[ec2-user@cafeserver ~]$
```

- To view its contents:

```
cat CafeDbDump.sql
```

-

```

[ec2-user@cafeserver ~]$ cat CafeDbDump.sql
-- MySQL dump 10.19  Distrib 10.2.38-MariaDB, for Linux (x86_64)
--
-- Host: localhost    Database: cafe_db
--
-- Server version      10.2.38-MariaDB
--
/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
/*!40101 SET NAMES utf8 */;
/*!40103 SET @OLD_TIME_ZONE=@@TIME_ZONE */;
/*!40103 SET TIME_ZONE='+00:00' */;
/*!40014 SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0 */;
/*!40014 SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0 */;
/*!40101 SET @OLD_SQL_MODE=@@SQL_MODE, SQL_MODE='NO_AUTO_VALUE_ON_ZERO' */;
/*!40111 SET @OLD_SQL_NOTES=@@SQL_NOTES, SQL_NOTES=0 */;
--
-- Current Database: `cafe_db`
--
CREATE DATABASE /*!32312 IF NOT EXISTS*/ `cafe_db` /*!40100 DEFAULT CHARACTER SET latin1 */;
USE `cafe_db`;
--
-- Table structure for table `order`
--
DROP TABLE IF EXISTS `order`;
/*!40101 SET @saved_cs_client      = @@character_set_client */;
/*!40101 SET character_set_client = utf8 */;
CREATE TABLE `order` (
  `order_number` int(5) NOT NULL AUTO INCREMENT,
  `order_date_time` timestamp NOT NULL DEFAULT current_timestamp(),
  `amount` decimal(10,2) NOT NULL DEFAULT 0.00,
  PRIMARY KEY (`order_number`)
) ENGINE=InnoDB AUTO INCREMENT=25 DEFAULT CHARSET=latin1;
/*!40101 SET character_set_client = @saved_cs_client */;

```

-  You have now successfully exported the existing database. In the next section, you'll import this [.sql](#) file into an Amazon RDS database.

## Step 4: Working with the RDS database

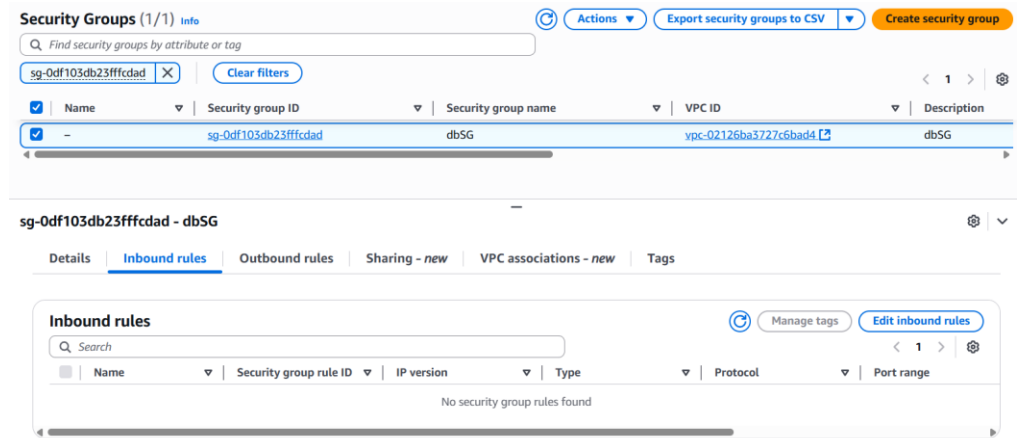
-  **Establishing a Network Connection to the RDS Instance**
  - In this task, we will initiate a connection from the EC2 instance to the new **Amazon RDS** instance to verify that it is accessible and ready for database import.
  -  **Step 1: Use the [mysql](#) command to connect**
    - In the EC2 terminal session, use the following command syntax to initiate the connection:

```
mysql -u admin -p --host <rds-endpoint>
```

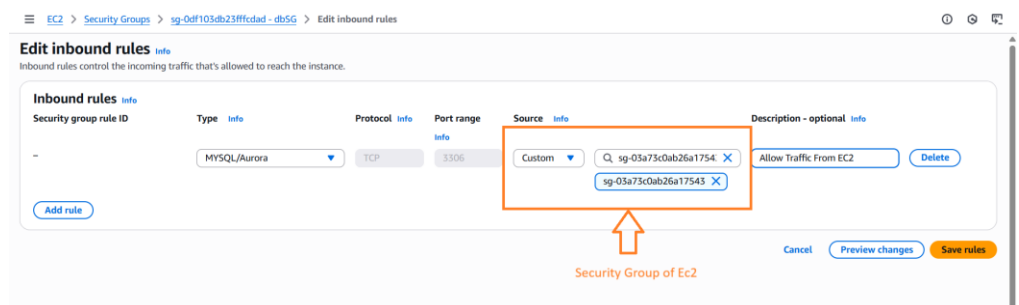
- An attempt was made to connect to the RDS database from the EC2 terminal, but the terminal became unresponsive and the connection was not established.

```
[ec2-user@cafeserver ~]$ mysql -u admin -p --host cafedatabase.cix4dckmcsqa.us-east-1.rds.amazonaws.com
Enter password:
```

- The issue was due to the RDS instance's security group having no inbound rules configured.



- To resolve the issue, an **inbound rule** was added to allow **MySQL/Aurora traffic (TCP port 3306)** from the **EC2 instance's security group**. This approach follows the principle of **security group chaining**, which enhances security and flexibility. Instead of allowing access from a wide IP range (such as `0.0.0.0/0` or a VPC CIDR block), the inbound rule on the RDS security group **specifically allows traffic only from the security group associated with the EC2 instance**.
  - This means:
    - Only instances that are part of the allowed security group (in this case, the EC2 instance) can access the RDS database.
    - No public IP or range is exposed.
    - This setup provides **tight control over who can access the database**, making the architecture more secure and scalable.



- After adding the rule, the connection from the EC2 terminal to the RDS database was attempted again.

```
[ec2-user@cafeserver ~]$ mysql -u admin -p --host cafedatabase.cix4dckmscqa.us-east-1.rds.amazonaws.com
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 81
Server version: 11.4.5-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> █
```

- 
- After successfully connecting to the RDS instance, run the following command:

```
show databases;
```

```
○
MariaDB [(none)]> show databases ;
+-----+
| Database                |
+-----+
| information_schema      |
| innodb                  |
| mysql                   |
| performance_schema     |
| sys                     |
+-----+
5 rows in set (0.01 sec)
```

- 
- Notice that the `cafe_db` database is **not listed yet**.  
This is expected behavior because **no data has been imported into the RDS instance at this point**.
- To disconnect from the MySQL session, simply run:

```
exit;
```

○

## New business requirement: Importing data and connecting the application to the new database

---

you exported the data from the database that the café application currently uses. You also established a network connection from the EC2 instance to the RDS instance. You can now work on the next business requirement.

you continue to take on the role of Sofia to import the cafe data into the RDS database instance. After you complete the import, you configure the application to use the new database.

### ☑ Step 5: Importing the data into the RDS database instance

---

- To import the data that was exported in Task 3 into your Amazon RDS database, use the following command. Be sure to replace `<rds-endpoint>` with the actual RDS endpoint:

```
mysql -u admin -p --host <rds-endpoint> < CafeDbDump.sql
```

- 
- When prompted, enter the **RDS instance password** that you defined during setup.
  - ☑ If the command runs without showing any errors, the data was likely imported successfully.

```
[ec2-user@cafeserver ~]$ mysql -u admin -p --host cafedatabase.cix4dckmscqa.us-east-1.rds.amazonaws.com < CafeDbDump.sql
Enter password:
[ec2-user@cafeserver ~]$
```

### 🔍 Verifying the Import

- To confirm that the data was imported correctly, connect again to the RDS instance:

```
[ec2-user@cafeserver ~]$ mysql -u admin -p --host cafedatabase.cix4dckmscqa.us-east-1.rds.amazonaws.com
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 81
Server version: 11.4.5-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

- 
- At the prompt, enter your password. Then run the following SQL commands:



```
show databases;

use cafe_db;

show tables;

select * from `order`;
```

```
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| cafe_db |
| information_schema |
| mysql |
| performance_schema |
| test |
+-----+
5 rows in set (0.00 sec)

MariaDB [(none)]> use cafe_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A
```

```
Database changed
MariaDB [cafe_db]> show tables;
+-----+
| Tables_in_cafe_db |
+-----+
| order |
| order_item |
| product |
| product_group |
+-----+
4 rows in set (0.00 sec)
```

```
MariaDB [cafe_db]> select * from `order`;
+-----+-----+-----+
| order_number | order_date_time | amount |
+-----+-----+-----+
| 1 | 2020-06-20 13:09:07 | 20.00 |
| 2 | 2020-06-21 13:09:23 | 24.00 |
| 3 | 2020-06-22 13:09:38 | 32.50 |
| 4 | 2020-06-22 13:09:49 | 10.50 |
| 5 | 2020-06-24 13:10:02 | 42.00 |
| 6 | 2020-06-25 13:10:20 | 14.00 |
| 7 | 2020-06-25 13:10:30 | 35.00 |
| 8 | 2020-06-27 13:10:36 | 6.00 |
| 9 | 2020-06-28 13:10:46 | 18.00 |
| 10 | 2020-07-02 13:10:58 | 18.00 |
| 11 | 2020-07-03 13:11:08 | 19.50 |
| 12 | 2020-07-04 13:11:17 | 9.00 |
| 13 | 2020-07-05 13:11:27 | 26.50 |
| 14 | 2020-07-05 13:11:40 | 42.00 |
| 15 | 2020-07-06 13:12:27 | 35.00 |
| 16 | 2020-07-09 13:12:35 | 19.50 |
| 17 | 2020-07-12 13:12:48 | 57.50 |
| 18 | 2020-07-14 13:13:04 | 34.00 |
| 19 | 2020-07-15 13:13:17 | 29.00 |
| 20 | 2020-07-18 13:13:27 | 14.00 |
| 21 | 2020-07-20 13:13:36 | 17.50 |
| 22 | 2020-07-21 13:13:47 | 33.50 |
| 23 | 2020-07-28 13:13:54 | 6.00 |
| 24 | 2020-07-28 13:14:07 | 35.00 |
| 25 | 2025-07-31 08:16:36 | 5.50 |
+-----+-----+-----+
25 rows in set (0.00 sec)
```

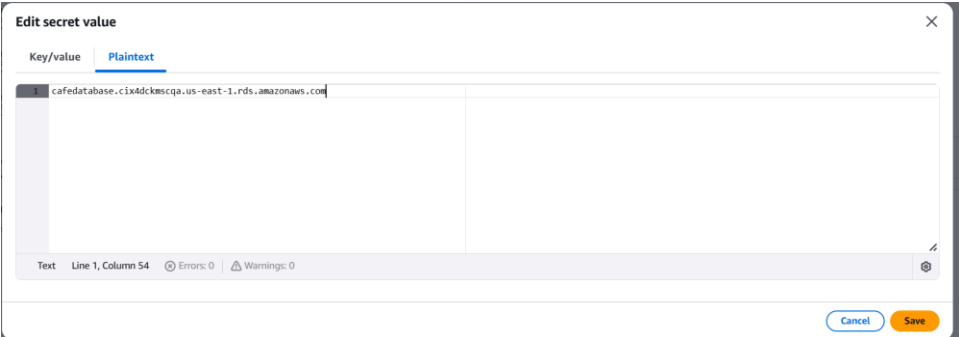
```
MariaDB [cafe_db]> 
```

## ☑ Step 6: Connecting the café to the new database

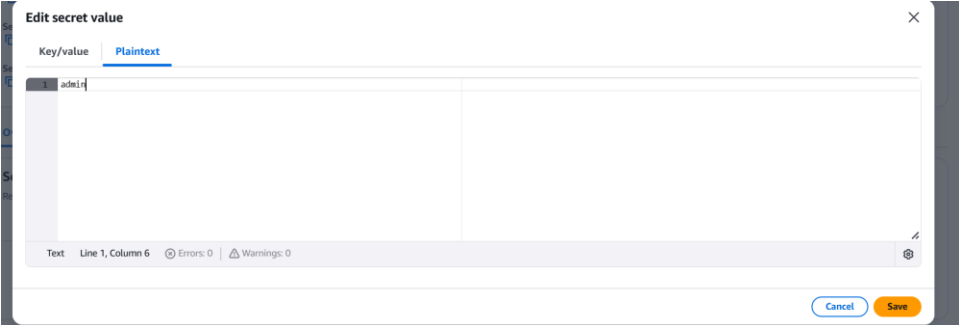
- In this task, you switch the café application to use the new **Amazon RDS** database instead of the old **MariaDB** database running on the **EC2 instance**.

-  **What to do:**

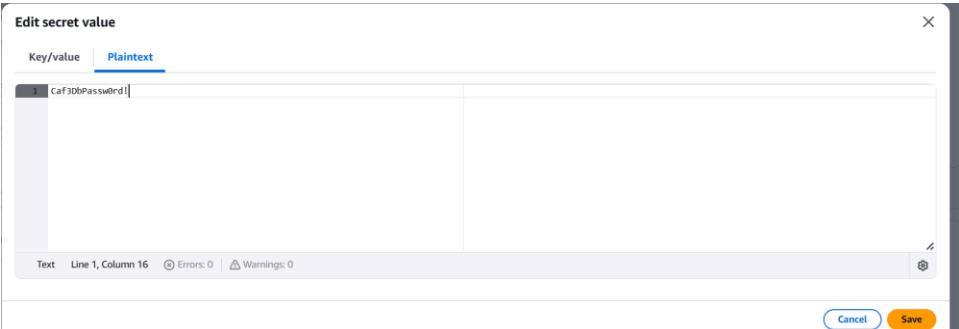
- Go to **Secrets Manager**.
- Locate the secret used by the café application.
- **Update the following values** inside the secret:
  - **dbUrl** → Replace with the **RDS endpoint**.



- **username** → Set to the RDS admin username.



- **password** → Set to the RDS admin password.



- Leave **dbName**, **currency**, **timeZone**, and **showServerInfo** unchanged.
- Save the secret changes.

- Why:

- The PHP code in the application is designed to read **database credentials from AWS Secrets Manager**. By updating the secret, the app will automatically connect to the new RDS instance without any changes in the code or network settings.

- Final Check:

- Stop the old local MariaDB server:

```
sudo service mariadb stop
```

```
(ec2-user@cafeserver ~)$ sudo service mariadb stop
Redirecting to /bin/systemctl stop mariadb.service
(ec2-user@cafeserver ~)$ service mariadb status
Redirecting to /bin/systemctl status mariadb.service
● mariadb.service - MariaDB 10.2 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; vendor preset: disabled)
   Drop-In: /usr/lib/systemd/system/mariadb.service.d
           -tokudb.conf
   Active: inactive (dead) since Thu 2025-07-31 12:21:00 UTC: 30s ago
   Process: 3716 ExecStartPost=/usr/libexec/mysql-check-upgrade (code=exited, status=0/SUCCESS)
   Process: 3645 ExecStart=/usr/libexec/mysqld --basedir=/usr $MYSQLD_OPTS $WSREP_NEW_CLUSTER (code=exited, status=0/SUCCESS)
   Process: 3511 ExecStartPre=/usr/libexec/mysql-prepare-db-dir %n (code=exited, status=0/SUCCESS)
   Process: 3487 ExecStartPre=/usr/libexec/mysql-check-socket (code=exited, status=0/SUCCESS)
   Main PID: 3645 (code=exited, status=0/SUCCESS)
   Status: "MariaDB server is down"

Jul 31 10:49:59 ip-10-0-0-21.ec2.internal mysql-check-upgrade[3716]: ERROR: ld.so: object '/usr/lib64/libjemalloc.so.1' from LD_PRELOAD cannot be preloaded (cann...ignore
Jul 31 10:49:59 ip-10-0-0-21.ec2.internal mysql-check-upgrade[3716]: ERROR: ld.so: object '/usr/lib64/libjemalloc.so.1' from LD_PRELOAD cannot be preloaded (cann...ignore
Jul 31 10:49:59 ip-10-0-0-21.ec2.internal mysql-check-upgrade[3716]: ERROR: ld.so: object '/usr/lib64/libjemalloc.so.1' from LD_PRELOAD cannot be preloaded (cann...ignore
Jul 31 10:49:59 ip-10-0-0-21.ec2.internal mysql-check-upgrade[3716]: ERROR: ld.so: object '/usr/lib64/libjemalloc.so.1' from LD_PRELOAD cannot be preloaded (cann...ignore
Jul 31 10:49:59 ip-10-0-0-21.ec2.internal mysql-check-upgrade[3716]: ERROR: ld.so: object '/usr/lib64/libjemalloc.so.1' from LD_PRELOAD cannot be preloaded (cann...ignore
Jul 31 10:49:59 ip-10-0-0-21.ec2.internal mysql-check-upgrade[3716]: ERROR: ld.so: object '/usr/lib64/libjemalloc.so.1' from LD_PRELOAD cannot be preloaded (cann...ignore
Jul 31 10:49:59 ip-10-0-0-21.ec2.internal systemd[1]: Started MariaDB 10.2 database server.
Jul 31 12:20:58 cafeserver systemd[1]: Stopping MariaDB 10.2 database server...
Jul 31 12:21:00 cafeserver systemd[1]: Stopped MariaDB 10.2 database server.
Hint: Some lines were ellipsized, use -l to show in full.
(ec2-user@cafeserver ~)$
```

- Open the app in a browser:

<http://<public-ip>/cafe/menu.php>

## Café

| Home                                                                                                                                                                                                         | Menu                                                                                                                                                                                               | Order History                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pastries                                                                                                                                                                                                     |                                                                                                                                                                                                    |                                                                                                                                                                                                                                          |
|  <p><b>Croissant</b><br/>\$1.50<br/>Fresh, buttery and fluffy... Simply delicious!<br/>Quantity: <input type="text"/></p> |  <p><b>Donut</b><br/>\$1.00<br/>We have more than half-a-dozen flavors!<br/>Quantity: <input type="text"/></p> |  <p><b>Chocolate Chip Cookie</b><br/>\$2.50<br/>Made with Swiss chocolate with a touch of Madagascar vanilla<br/>Quantity: <input type="text"/></p> |
|                                                                                                                           |                                                                                                                |                                                                                                                                                     |

- Place a new order and go to **Order History**.

HomeMenuOrder History

Pastries



Croissant

\$1.50

Fresh, buttery and fluffy... Simply delicious!

Quantity:



Donut

\$1.00

We have more than half-a-dozen flavors!

Quantity:



Chocolate Chip Cookie

\$2.50

Made with Swiss chocolate with a touch of Madagascar vanilla

Quantity:



Muffin

\$3.00

Banana bread, blueberry, cranberry or apple



Strawberry Blueberry Tart

\$3.50

Bursting with the taste and aroma of fresh fruit



Strawberry Tart

\$3.50

Made with fresh ripe strawberries and a delicious whipped cream



Coffee

\$3.00

Freshly-ground black or blended Columbian coffee

Quantity:



Hot Chocolate

\$3.00

Rich and creamy, and made with real chocolate

Quantity:



Latte

\$3.50

Offered hot or cold and in various delicious flavors

Quantity:

Order Total: \$6.50

Submit Order

# Café

HomeMenuOrder History

## Order Confirmation

Thank for your order! It will be available for pickup within 15 minutes. Your order number and details are shown below.

Order Number: 25    Date: 2025-07-31    Time: 12:16:06    Total Amount: \$6.50

| Item                  | Price  | Quantity | Amount |
|-----------------------|--------|----------|--------|
| Croissant             | \$1.50 | 2        | \$3.00 |
| Donut                 | \$1.00 | 1        | \$1.00 |
| Chocolate Chip Cookie | \$2.50 | 1        | \$2.50 |

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# Café

| Home | Menu | Order History |
|------|------|---------------|
|------|------|---------------|

Order Number: 25

Date: 2025-07-31

Order History

Time: 12:16:06 Total Amount: \$6.50



| Item                  |
|-----------------------|
| Croissant             |
| Donut                 |
| Chocolate Chip Cookie |

| Price  | Quantity | Amount |
|--------|----------|--------|
| \$1.50 | 2        | \$3.00 |
| \$1.00 | 1        | \$1.00 |
| \$2.50 | 1        | \$2.50 |

Order Number: 24 Date: 2020-07-28 Time: 13:14:07 Total Amount: \$35.00

| Item                      | Price  | Quantity | Amount  |
|---------------------------|--------|----------|---------|
| Strawberry Blueberry Tart | \$3.50 | 4        | \$14.00 |
| Strawberry Tart           | \$3.50 | 3        | \$10.50 |
| Latte                     | \$3.50 | 3        | \$10.50 |

Order Number: 23 Date: 2020-07-28 Time: 13:13:54 Total Amount: \$6.00

| Item   | Price  | Quantity | Amount |
|--------|--------|----------|--------|
| Coffee | \$3.00 | 2        | \$6.00 |

Order Number: 22 Date: 2020-07-21 Time: 13:13:47 Total Amount: \$33.50

| Item                  | Price  | Quantity | Amount |
|-----------------------|--------|----------|--------|
| Chocolate Chip Cookie | \$2.50 | 3        | \$7.50 |

- 
- If previous and new orders are displayed, it confirms the application is successfully connected to the RDS database.
-

```
MariaDB [cafe_db]> select * from `order`;
```

| order_number | order_date_time     | amount |
|--------------|---------------------|--------|
| 1            | 2020-06-20 13:09:07 | 20.00  |
| 2            | 2020-06-21 13:09:23 | 24.00  |
| 3            | 2020-06-22 13:09:38 | 32.50  |
| 4            | 2020-06-22 13:09:49 | 10.50  |
| 5            | 2020-06-24 13:10:02 | 42.00  |
| 6            | 2020-06-25 13:10:20 | 14.00  |
| 7            | 2020-06-25 13:10:30 | 35.00  |
| 8            | 2020-06-27 13:10:36 | 6.00   |
| 9            | 2020-06-28 13:10:46 | 18.00  |
| 10           | 2020-07-02 13:10:58 | 18.00  |
| 11           | 2020-07-03 13:11:08 | 19.50  |
| 12           | 2020-07-04 13:11:17 | 9.00   |
| 13           | 2020-07-05 13:11:27 | 26.50  |
| 14           | 2020-07-05 13:11:40 | 42.00  |
| 15           | 2020-07-06 13:12:27 | 35.00  |
| 16           | 2020-07-09 13:12:35 | 19.50  |
| 17           | 2020-07-12 13:12:48 | 57.50  |
| 18           | 2020-07-14 13:13:04 | 34.00  |
| 19           | 2020-07-15 13:13:17 | 29.00  |
| 20           | 2020-07-18 13:13:27 | 14.00  |
| 21           | 2020-07-20 13:13:36 | 17.50  |
| 22           | 2020-07-21 13:13:47 | 33.50  |
| 23           | 2020-07-28 13:13:54 | 6.00   |
| 24           | 2020-07-28 13:14:07 | 25.00  |
| 25           | 2025-07-31 12:16:06 | 6.50   |

25 rows in set (0.00 sec)

## Update from the café

Everyone at the café is happy with the results of the database migration. Sofia and Nikhil now have more free time on weekends, which means that Frank and Martha are saving money on labor costs.

Sofia takes a minute to relax with her friends. However, she's already thinking about improvements. A good next step would be to reduce the size of the Amazon Elastic Block Store (Amazon EBS) volume that the EC2 instance uses. They could also save on expenses by changing the EC2 instance type to a smaller size. Because the database no longer runs on the EC2 instance, the instance now has extra hard drive space, and it might also not need as many CPU and memory resources.

## Conclusion

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In this lab, we successfully migrated the café application's database from a local MariaDB instance on EC2 to a managed Amazon RDS database. We exported and imported the data, updated the application's connection settings using AWS Secrets Manager, and verified that the application continued to function seamlessly after the migration. This process demonstrates the benefits of using managed services like Amazon RDS for improved scalability, reliability, and operational efficiency in modern cloud-based applications.

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